Masking Graph Cross-Convolution Network for Multispectral Point Cloud Classification

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***Abstract*— Achieving accurate 3-D environment perception is a key task in the field of remote sensing. Multispectral point cloud has rich integrated 3-D spatial–spectral information, which provides a data basis for realizing more detailed scene understanding and perception. However, the diversity of land covers and the complexity of its features pose challenges to classification. In addition, the current methods mechanically pool and fuse local features to obtain global information, which has limited the utility for multispectral point cloud classification. In this article, we propose a masking graph cross-convolution network (MGC2N), which aims to address these problems by utilizing spectral features to construct point-to-point relationships independent of spatial distance. A self-attention masking (SAM) module and a spatial–spectral cross-convolution (S2C2) module are innovatively designed into the proposed MGC2N. The former is used to adaptively adjust the nodes and edges of the adjacency matrix to dynamically extract effective features for different land covers; the latter is used to extract spatial distribution features and local spectral features of the land covers in the scene to enhance the discriminative ability of the learned features. Our method achieves the best-in-class performance on two real multispectral point cloud datasets, demonstrating its effectiveness in improving classification accuracy and robustness.**

***Index Terms*— Graph neural network (GNN), multispectral point cloud, self-attention masking (SAM), spatial–spectral cross-convolution (S2C2).**

I. Introduction

THREE-DIMENSIONAL point clouds are widely used in various fields because of their rich geometric structure information [1], [2]. Multispectral/hyperspectral images, on the other hand, are widely used in various detection and monitoring tasks due to their rich spectral information [3], [4], [5], [6]. In the field of remote sensing, there are scholars who combine multispectral/hyperspectral imagery andairborne LiDAR point clouds to achieve more accurate and detailed 3-D environment perception [7], [8]. However, difficulties in data alignment efforts limit the potential of combining spatial and spectral information. The emergence of multispectral point clouds, which naturally carry rich spatial and spectral information, has received extensive attention from remote sensing scholars [9], [10]. Classification of multispectral point clouds still faces many challenges due to the wide range of coverage scenarios, diversity of land cover, and complexity of features.

In recent years, the field of multispectral point cloud land cover classification has experienced a significant transformation, driven by the rapid advancements in deep learning methods. These methods have proven highly effective for learning and extracting multilayered features from multispectral point clouds. Various methods to processing point clouds have been developed, ranging from rasterized image-based methods [11], [12], [13] to point-based multilayer perceptron (MLP) methods such as those presented in [14], [15], and [16] and extending to graph convolutional network (GCN) methods [17], [18], [19], [20]. While these methodologies have marked some successes in classifying multispectral point clouds, they commonly struggle with retaining geometric contour information of the land covers, often resulting in obscured and confusing boundaries. This issue becomes particularly pronounced when two land covers share similar spatial structures and spectral features. We believe that this problem stems from the following reasons. First, the complexity and diversity of the land covers in multispectral point clouds imply that a unified and fixed feature extraction approach lacks flexibility and cannot be efficiently applied to all land covers. Second, the traditional methods of aggregating local features to capture global information are of limited use for large-scale point clouds. In addition, the feature extraction process often reuses complex and redundant information from multispectral point clouds, leading to a high degree of similarity in high-level semantic features, which in turn affects the performance of the network.

To address these problems, we propose the masking graph cross-convolution network (MGC2N), which consists of a self-attention masking (SAM) module and a spatial–spectral cross-convolution (S2C2) module. Our main contributions are as follows.

Received 4 August 2024; revised 8 December 2024; accepted 20 February 2025. Date of publication 26 February 2025; date of current version 12 March 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62201237; in part by Yunnan Fundamental Research Projects under Grant 202401AW070019, Grant 202101BE070001-008, and Grant 202301AV070003; and in part by the Major Science and Technology Projects in Yunnan Province under Grant 202302AG050009. *(Corresponding author: Tao Shen.)*

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Digital Object Identifier 10.1109/TGRS.2025.3545783

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1) We designed an SAM module that adaptively adjusts the nodes and edge relationships of the adjacency matrix to provide distinct contextual information for each point.
2) We developed an S2C2 module, which uses cross-convolution to simultaneously extract global spatial distribution features and local spectral features of land covers. This design enhances the ability to distinguish between different land covers and captures geometric features while also reducing the similarity between high-level semantic features.
3) Based on the above two points, we propose MGC2N, which perceives more detailed features and achieves accurate multispectral point cloud classification. We achieve state-of-the-art (SOTA) classification performance on two real multispectral point cloud datasets.

The rest of this article is organized as follows. Section II summarizes the related work. Section III describes our network and method in detail. Section IV describes extensive comparison and ablation experiments performed on two real multispectral point cloud datasets to demonstrate the effectiveness of our method. Finally, Section V concludes this article.

II. Related Work

A. Multispectral Point Cloud Classification

In recent years, multispectral point cloud classification methods are mainly categorized into two types: image-based methods [21], [22], [23] and point-based methods [24], [25], [26]. Image-based methods rasterize the multispectral point clouds into the 2-D image and then perform feature extraction on the 2-D image to achieve point clouds classification. Point-based methods, on the other hand, directly perform feature extraction on the original point clouds. In the early days, both types of methods relied on manually designed feature descriptors for feature extraction. However, with the development of deep learning, many studies have turned to deep neural network frameworks for feature extraction.

Due to the unstructured and disordered nature of point cloud, it is difficult to directly use traditional convolutional neural networks (CNNs) to process 3-D point cloud. Therefore, the researchers first introduced deep learning methods in the feature extraction process of rasterized images. Pan et al. [27] first achieved significant results by replacing traditional machine learning classification methods with stacked MLPs. Subsequently, they designed a CNN model for extracting local features from rasterized images for accurate land cover classification [12]. To enhance the feature extraction capability, Yu et al. [11] proposed a hybrid capsule network (HCapsNet) with a dual-stream structure to extract global and local features respectively. Similarly, in order to learn contextual semantic features at long distances, Yu et al. [13] introduced the Transformer structure and proposed CapViT. Although these methods have achieved satisfactory results, the loss of spatial information from rasterized images is unavoidable [28], [29], [30].

Some researchers have developed deep learning methods to extract features directly from multispectral point clouds. Inspired by PointNet++ [15], Jing et al. [16] proposed a point-based MLP method, SE-PointNet++, which adds squeezing and excitation modules to PointNet++, enhances the network's ability to extract local features and utilizes multichannel spectral information to improve the network's classification performance. However, point-based MLP methods mainly capture point-by-point features or point-set features, thus lacking the ability to capture the local fine-grained features. The Transformer framework has garnered significant attention due to its enhanced ability to capture contextual semantic features through the introduction of the attention mechanism [31], [32]. Xiao et al. [33] integrated a deformable attention mechanism into the Transformer and combined it with CNN to improve the learning of multi-scale representations in high-resolution remote sensing images. MIA-Net [34] utilizes local trigonometric encoding, feature sampling, and intuitive attention mechanisms to achieve SOTA accuracy and computational efficiency in airborne point cloud segmentation. However, the substantial computational burden associated with the Transformer remains a significant challenge.

In contrast, graph neural networks (GNNs) excel in the field of point cloud processing, which can efficiently deal with disordered data by constructing graphs to aggregate key information. Zhao et al. [36] utilized GCN [35] to obtain global contextual features of multispectral point clouds by revealing spatial relationships between points. On this basis, graph attention convolutional networks (GACNets) [37] dynamically adjust the kernel shape by introducing a self-attention mechanism to enhance object delineation and reduce segmentation errors for point clouds. graph convolutional networks via initial residual and identity mapping (GCNII) [38], on the other hand, solves the oversmoothing problem in GNNs by introducing initial residual connections and constant mapping, thus improving the performance of deep GCNs. Similarly, graph convolutional generalized network (GCBNet) [39] combines graph convolutional layers with stacked regular convolutional layers, thus improving the network's ability to extract deep features. An increasing number of researchers are focusing on developing GNN methods for multispectral point classification. Yang et al. [40] proposed MS-AMCNN to enhance the recognition of land covers by constructing multiscale local spatial graphs to extract multilevel local features. However, these methods emphasized the application of spatial relationships between points and neglected the relationships in the spectral feature dimension. Wang et al. [18] proposed deep spatial graph convolution network with adaptive spectral aggregated residuals (DSGCN-ASR), which discussed the role of spatial and spectral relationships between points and improved the performance of the network by enhancing the utilization of spectral features. In their follow-up work, they mined the higher order topological relationships between points through multicore mapping and achieved remarkable results by dynamically selecting topological relationships [19]. Graph-based methods are becoming a mainstream approach for processing multispectral point cloud.

B. Masking Strategy

In the field of deep learning, masking techniques have been widely studied and validated, primarily for highlighting impor-

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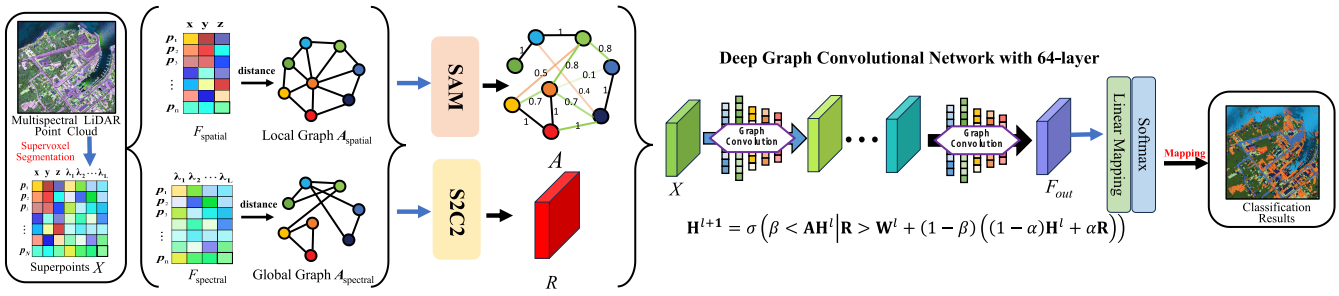


Fig. 1. Framework of MGC2N.

tant features and suppressing irrelevant information to improve model performance. In natural language processing (NLP), masking techniques enhance model training by concealing parts of the input. For instance, the bidirectional encoder representations from transformers (BERT) model randomly masks some input text and trains the model to predict the masked words, thereby learning rich contextual semantic information [41]. This significantly improves the model's performance across various downstream tasks, including question answering [42], text classification [43], and machine translation [44].

Similarly, in image processing, masking techniques have also yielded significant results. Early models such as U-Net incorporated masks in their encoder–decoder structures to achieve high-precision segmentation of medical images [45]. Subsequently, Mask R-CNN [46] combined region proposal networks with mask segmentation, enabling precise instance-level object segmentation. Masking strategies are also effective and widely used in applications such as scene understanding of 3-D point clouds [47] and land cover classification of remote sensing images [48].

III. Methodology

In this article, we propose a novel multispectral point cloud classification method based on GCN, named MGC2N. This method can realize large-scale multispectral point cloud classification in remote sensing scenes. As shown in Fig. 1, it mainly consists of two modules: SAM and S2C2.

A. Network Architecture

Due to the sheer size of multispectral point clouds in remote sensing scenes, which often consist of millions of points, most existing networks and methods struggle to directly process such data. Therefore, MGC2N avoids direct input of raw points into the network. Specifically, we first segment the millions of original point clouds into thousands of superpoints using supervoxel segmentation [49] and then classify these superpoints. The classification results of the superpoints are then mapped back to the original point clouds to achieve the classification of the original points. Specifically, each superpoint has a unique number, while each primitive point is labeled with the number of its corresponding superpoint. Using this correspondence, we can realize the mapping of labels between superpoints and original points. This method reduces the demand for computational resources and improves inference speed.

The superpoints can be denoted as $X = [x, y, z, \lambda_1, \ldots, \lambda_L] \in \mathbb{R}^{(L+3)\times N}$, where L is the number of spectral bands and N is the number of superpoints. We denote the 3-D spatial information (i.e., $[x, y, z]$) and spectral features (i.e., $[\lambda_1 \cdots \lambda_L]$) by F_{spatial} and F_{spectral}, respectively.

The MGC2N leverages the rich spatial and spectral information to construct spatial local adjacency graphs and global spectral connectivity graphs, respectively. The local graph $\boldsymbol{A}_{\text{spatial}}$ is constructed based on F_{spatial}, which involves establishing connections between points that are spatially adjacent. Conversely, the global graph $\boldsymbol{A}_{\text{spectral}}$ is derived from F_{spectral}, focusing on establishing relationships between points that are spectrally homogeneous, regardless of spatial distance. Subsequently, $\boldsymbol{A}_{\text{spatial}}$ and $\boldsymbol{A}_{\text{spectral}}$ are processed within the SAM module for optimization and fusion, resulting in a new graph $\boldsymbol{A}$. This new graph, along with X, serves as input to the backbone network. Meanwhile, S2C2 extracts discriminative features of the land cover and incorporates them into the backbone network as initial residuals and identity mappings. Following the softmax layer, the classification results for the superpoints are obtained. Ultimately, these results are mapped back to the original point cloud for statistical analysis.

B. Self-Attention Masking

In order to provide the land covers with adaptive receptive fields to facilitate the extraction of discriminative contextual information, we introduce a masking strategy. As shown in Fig. 2, we use masks to adjust and fuse graphs ($\boldsymbol{A}_{\text{spatial}}$ and $\boldsymbol{A}_{\text{spectral}}$). Specifically, we use two complementary mask matrices ($\boldsymbol{M}_A$ and $\boldsymbol{M}_B$) to perform the Hadamard product with $\boldsymbol{A}_{\text{spatial}}$ and $\boldsymbol{A}_{\text{spectral}}$, respectively. The parameters in both $\boldsymbol{M}_A$ and $\boldsymbol{M}_B$ are automatically optimized and updated, where $\boldsymbol{M}_A$ is automatically optimized based on the network loss and $\boldsymbol{M}_B$ is varied according to the changes in $\boldsymbol{M}_A$. In this way, we are able to dynamically regulate each connection in the topological graphs and adaptively learn the importance of the connections between each point. Furthermore, we fuse the optimized graphs in an additive manner to obtain a new graph $\boldsymbol{A}$. Since the spectral graph is independent of spatial distance, the graph $\boldsymbol{A}$ enables the aggregation of different distance information. The formula is as follows:

$$\begin{aligned}\boldsymbol{A} &= \boldsymbol{A}_{\text{spatial}} * \boldsymbol{M}_A + \boldsymbol{A}_{\text{spectral}} * (\mathbf{J} - \boldsymbol{M}_A) \\ &= \boldsymbol{A}_{\text{spatial}} * \boldsymbol{M}_A + \boldsymbol{A}_{\text{spectral}} * \boldsymbol{M}_B \end{aligned} \tag{1}$$

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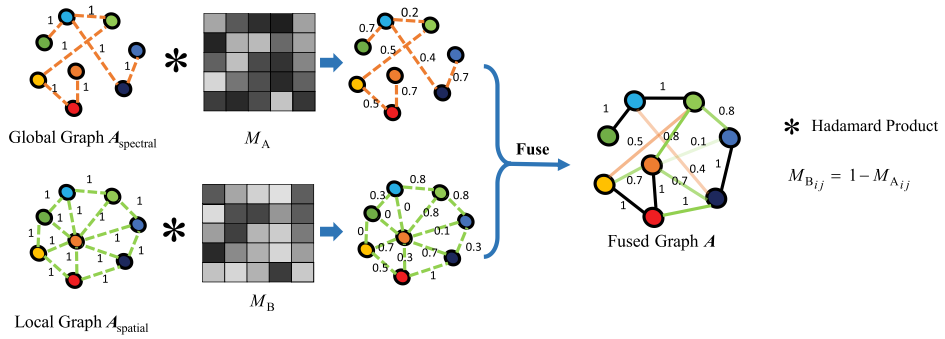


Fig. 2. SAM module.

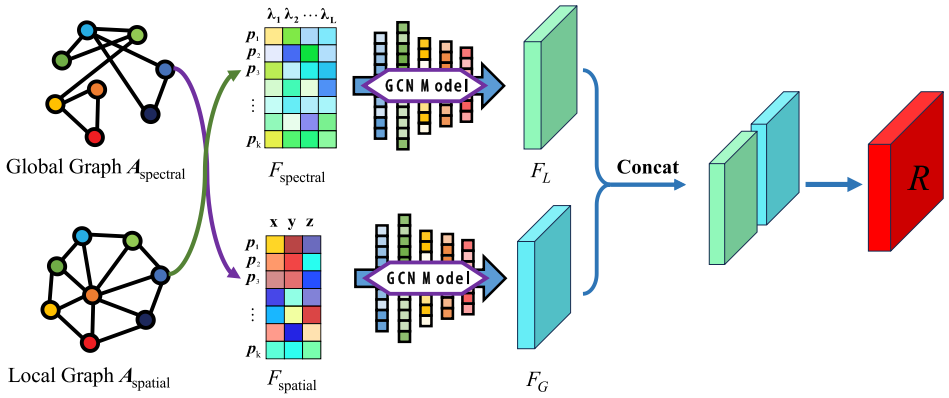


Fig. 3. S2C2 module.

where $*$ represents the Hadamard product and $\mathbf{J}$ represents ones matrix.

With the SAM module, we can provide more effective contextual information for each point and enhance the network's ability to sense important features. However, since the dimension of the masks is the same as the number of input superpoints, which is $N \times N$, a shallow network cannot effectively learn the requisite parameters due to the large parameter space. Therefore, it is necessary to introduce a deep network as the backbone to adequately fit the model.

C. Spatial–Spectral Cross-Convolution

In order to learn the global and local features more efficiently and enhance the feature representation of the point clouds, we designed an S2C2 module, as shown in Fig. 3. We aim to leverage 1-D information to establish topological relations that integrate information from the other dimension, enabling the extraction of higher order semantic features in spatial–spectral data. Specifically, we input F_{spatial} and $\boldsymbol{A}_{\text{spectral}}$ into a single-layer GCN to learn the global spatial distribution features F_G of point clouds. Simultaneously, we input F_{spectral} and $\boldsymbol{A}_{\text{spatial}}$ into another single-layer GCN to obtain local spectral features F_L of point clouds. We aim to enhance the distinction of land covers by incorporating spatial constraints through the extraction of F_G. Similarly, we aim to introduce local spectral constraints by extracting F_L, thereby enhancing the recognition of the geometric structure of the land cover. In order to facilitate the utilization of these features, we integrated F_G and F_L. The feature propagation process for the single-layer GCN and the feature fusion process are as follows:

$$F_G = \sigma\left(\boldsymbol{A}_{\text{spectral}} F_{\text{spatial}} W_1\right) \tag{2}$$

$$F_L = \sigma\left(\boldsymbol{A}_{\text{spatial}} F_{\text{spectral}} W_2\right) \tag{3}$$

$$R = F_G | F_L \tag{4}$$

where σ is the ReLU activation function, W_1 and W_2 are the trainable weight parameter, $|$ is the concatenate operation, and R is the fused features.

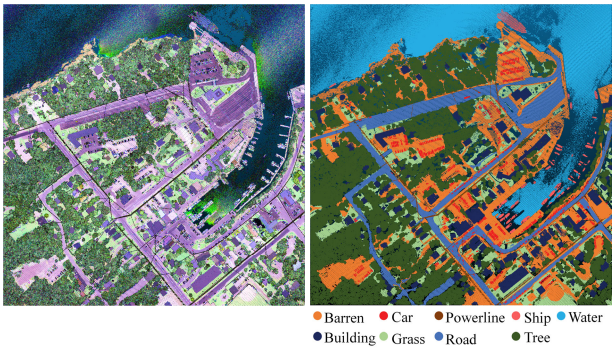
With the S2C2 module, we extract higher order semantic features through the interaction of spatial and spectral information, thereby improving the recognition of land covers. Notably, the cross-convolution technique also reduces information redundancy and the similarity between higher order semantic features.

D. Feature Propagation Process and Loss Function

The backbone network is a 64-layer deep GCN, and notably, its first layer is different from the others, with an extra graph convolutional layer to obtain features $\mathbf{H}^0$. The features X of the superpoints are used as the initial input, while the topological graph $\boldsymbol{A}$ and the fusion features $\mathbf{R}$ are input to each layer of the backbone to help feature propagation and learning. The specific process is as follows:

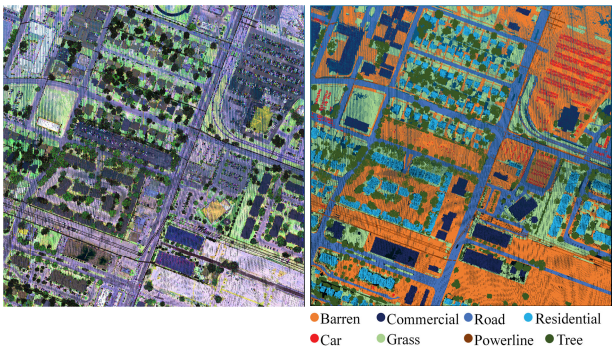
$$\mathbf{H}^0 = \sigma\left(\boldsymbol{A} X_{N\times(3+L)} \mathbf{W}^0\right) \tag{5}$$

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(a) (b)

Fig. 4. Visualization and labeling of HT dataset. (a) Visualization of the original point clouds. (b) Manually labeled ground truth.



(a) (b)

Fig. 5. Visualization and labeling of UH dataset. (a) Visualization of the original point clouds. (b) Manually labeled ground truth.

$$\mathbf{H}^{l+1} = \sigma\left(\beta\left\langle A\mathbf{H}^{l}\middle|\mathbf{R}\right\rangle > \mathbf{W}^{l} + (1-\beta)\left((1-\alpha)\mathbf{H}^{l} + \alpha\mathbf{R}\right)\right) \tag{6}$$

where $\mathbf{H}$ stands for intermediate features and l for the number of layers, and α and β are parameters used to control the influence of R. α is a trainable parameter, while β is a coefficient that decays with the number of network layers as follows:

$$\beta = \ln(1/2l + 1). \tag{7}$$

Finally, the fully connected and softmax layers are utilized to obtain the results for each superpoint as follows:

$$\hat{y} = \text{Softmax}(\text{Linear}(F_{N\times d}, F_{N\times c})) \tag{8}$$

where $\hat{y}$ denotes the prediction result, d denotes the dimension of the output feature in the last layer, and c denotes the number of classes.

In the network, we utilize the cross-entropy loss function to train the network, which is represented as follows:

$$\mathcal{L}\text{oss} = -\sum_{i=1}^{m}\sum_{j=1}^{C} y_{ij}\log\left(\hat{y}_{ij}\right) \tag{9}$$

where m is the number of superpoints in the dataset participating in the training, C is the number of classes, y_{ij} is the ground truth of the ith superpoint belonging to the jth class, and $\hat{y}_{ij}$ is the predicted probability of the ith superpoint belonging to the jth class.

Algorithm 1 MGC2N Training

Input: Superpoints $X = [x, y, z, \lambda_1 \ldots \lambda_L] \in \mathbb{R}^{6\times N}$; the labels of superpoints $Y = \{y^1, \ldots, y^n\}$.

Output: Predicted labels of superpoints.

Max-min normalization X, $x_i = \frac{x_i - \min(X)}{\max(X) - \min(X)}$;

Splitting spatial and spectral features:
$F_{\text{spectral}} = [x, y, z] \in \mathbb{R}^{3\times N}$;
$F_{\text{spatial}} = [\lambda_1 \cdots \lambda_L] \in \mathbb{R}^{3\times N}$;

Calculate the initial adjacency matrix $\boldsymbol{A}_{\text{spatial}}$ and $\boldsymbol{A}_{\text{spectral}}$:

$$\boldsymbol{A}_{\text{spatial}_{ij}} = \begin{cases} 1, & \text{if } \exp(\left\|x_i - x_j\right\|^2_{\text{spatial}}) - 1 \le 0.1 \\ 0, & \text{otherwise} \end{cases};$$

$$\boldsymbol{A}_{\text{spectral}_{ij}} = \begin{cases} 1, & \text{if } \exp(\left\|x_i - x_j\right\|^2_{\text{spectral}}) - 1 \le 0.1 \\ 0, & \text{otherwise} \end{cases};$$

for i in epochs **do**
  $A = SAM(\boldsymbol{A}_{\text{spectral}}, \boldsymbol{A}_{\text{spatial}})$
  $R = S2C2(\boldsymbol{A}_{\text{spectral}}, F_{\text{spatial}}, \boldsymbol{A}_{\text{spatial}}, F_{\text{spectral}})$;
  $F_{out} = GCN_{deep}(\boldsymbol{A}, X, R, \beta = 0.5)$;
  $\hat{y} = \text{Softmax}(\text{Linear}(F_{out}))$;
  Calculate loss $\mathcal{L}oss = CrossEntropy(y, \hat{y})$;
  loss.backward();
  model.update().
end for

Algorithm 1 shows the training process of our proposed MGC2N.

IV. Experiments

All experiments were conducted on a system equipped with an Intel Core i5-12600KF CPU @ 3.70 GHz and an NVIDIA GeForce ray tracing texel extreme (RTX) 3060 graphics processing unit (GPU), featuring 12 GB of memory. We trained the model using the Adam optimizer with an initial learning rate of 0.01 and employed a cyclical learning rate strategy, completing six cosine cycles per training session. The hyperparameter β was set to 0.5. To quantitatively assess the performance of our proposed network, we employed five commonly used metrics: accuracy, precision, recall, F1-score, and intersection over union (IoU). To validate the effectiveness of our method, we conducted comparative analyses with SOTA graph methods, including GCN [35], GCNII [38], GACNet [37], GCBNet [39], DSGCN-ASR [18], and multikernel graph structure learning (MKGSL) [19] on two datasets. Finally, we performed ablation experiments on the two main modules in the network, SAM and S2C2.

A. Description of the Data

In this article, we utilize two datasets: the Harbor of Tobermory (HT), depicted in Fig. 4, and the University of Houston (UH), illustrated in Fig. 5. Both were acquired using an airborne multispectral LiDAR system, the Optech Titan, which features three laser channels capable of capturing intensity information in three bands (1550, 1064, and 532 nm).

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TABLE I
COMPARISON OF EVALUATION METRICS (%) ON THE HT DATASET

Method	Metrics	Barren	Building	Car	Grass	Powerline	Road	Ship	Tree	Water	Macro Average	OA
GCN [35]	Precision	**75.50**	72.88	33.42	87.70	66.37	82.38	57.34	83.85	91.24	72.30	81.36
	Recall	82.45	69.97	20.24	81.09	4.89	**70.57**	34.61	99.20	88.85	61.32	
	F1-score	78.82	71.39	25.21	84.27	9.11	76.02	43.16	90.88	90.03	66.36	
	IoU	65.04	55.51	14.42	72.81	4.77	61.31	27.52	83.29	81.86	51.84	
GCNII [38]	Precision	71.42	79.15	38.97	88.60	**82.17**	88.57	**82.58**	89.35	91.71	79.17	84.83
	Recall	85.88	77.71	28.40	88.44	9.12	67.80	31.06	**99.40**	94.29	64.68	
	F1-score	77.99	78.42	32.86	88.52	16.42	76.80	45.14	94.11	92.98	71.19	
	IoU	63.92	64.50	19.66	79.41	8.94	62.34	29.15	88.87	86.89	55.96	
GACNet [37]	Precision	72.37	71.53	29.14	88.46	51.50	79.46	70.75	92.60	89.59	71.71	84.77
	Recall	79.67	69.64	18.86	82.95	21.48	67.78	35.53	98.98	83.46	62.04	
	F1-score	75.85	70.57	22.90	85.62	30.32	73.16	47.31	95.68	86.42	66.53	
	IoU	61.09	54.53	12.93	74.85	17.87	57.67	30.98	91.72	76.08	53.08	
GCBNet [39]	Precision	62.48	**90.04**	42.23	88.37	69.87	91.39	75.02	90.54	90.63	77.84	84.70
	Recall	85.51	69.97	32.96	81.76	41.27	58.92	32.42	99.33	97.02	66.58	
	F1-score	72.21	78.75	37.02	84.94	51.89	71.65	45.27	94.73	93.71	71.77	
	IoU	56.50	64.95	22.72	73.82	35.04	55.83	29.26	89.99	88.17	57.36	
DSGCN-ASR [18]	Precision	72.18	88.36	29.28	90.95	69.95	77.08	47.02	95.41	97.86	74.23	87.57
	Recall	78.40	78.74	38.73	86.34	33.12	64.81	48.64	99.05	97.24	69.45	
	F1-score	75.16	83.27	33.35	88.59	44.96	70.41	47.81	97.19	97.55	71.76	
	IoU	60.21	71.34	20.01	79.51	29.00	54.34	31.42	94.54	95.21	59.51	
MKGSL [19]	Precision	68.30	82.48	38.24	84.97	45.70	**92.74**	37.47	**97.77**	97.79	71.72	88.68
	Recall	81.74	**85.78**	**58.78**	90.83	**57.14**	62.46	68.14	98.03	**99.35**	**78.03**	
	F1-score	74.42	84.10	**46.34**	87.81	50.78	74.65	48.35	97.90	**98.57**	74.74	
	IoU	59.26	72.56	**30.16**	78.26	34.03	59.55	31.88	95.89	**97.17**	62.08	
MGC2N(Ours)	Precision	73.53	86.05	**43.09**	**91.25**	79.69	92.28	68.93	96.86	**98.38**	**81.12**	**90.30**
	Recall	**88.00**	84.09	43.32	**91.39**	39.99	68.06	**70.97**	99.04	98.61	75.94	
	F1-score	**80.12**	**85.06**	43.20	**91.32**	**53.25**	**78.34**	**69.93**	**97.94**	98.49	**78.44**	
	IoU	**68.34**	**74.00**	27.55	**84.03**	**36.29**	**64.39**	**53.77**	**95.96**	97.03	**66.65**	

Due to the different downward slopes of these channels, not every point contains intensity information in all three bands. Therefore, we use the 1550-nm band as the core band and employ the nearest search method to incorporate intensity information from the other two bands into the core band.

The HT dataset contains 7 181 982 points and is manually labeled into nine classes of land cover: barren, building, car, grass, powerline, road, ship, tree, and water. The UH dataset contains 4 436 470 points and is manually labeled into eight classes of land cover: barren, commercial building, car, grass, powerline, road, residential building, and tree. It is important to note that in both datasets, the car and powerline classes have significantly fewer instances compared to other classes.

Due to a large number of point clouds, which cannot be directly processed by our equipment, we use the method in [49] to segment them into superpoints. The HT dataset is segmented into 9606 superpoints, and the UH dataset is segmented into 9350 superpoints. Since the labels of the superpoints are generated by voting on the points contained within them, some errors arise when mapping these labels back to the original point clouds. We accounted for these errors in our experiments. Consequently, for visualization purposes, we map the superpoint labels generated after voting back to the original point clouds and use these mapped labels as the ground truth. However, for calculating the evaluation metrics, we still rely on the manually labeled ground truth.

B. Performance and Analysis of Experiments

To ensure experimental fairness, we constructed topological graphs between points as follows: GACNet, DSGCN-ASR, and MKGSL each using their own methods to build adjacency graphs. In contrast, other methods such as GCN, GCNII, and GCBNet uniformly use a spatial and spectral graph averaged weighted fusion graph as the final adjacency graph.

1) Classification Results on HT: Table I presents the classification metrics for each land cover type and overall evaluation metrics. It is evident from the table that our method achieves a macro F1-score (MF) of 78.44%, a macro IoU (MIoU) of 66.65%, and an overall accuracy (OA) of 90.30%, significantly enhancing the classification of various land covers. Notably, the MIoU has improved by 4.57%, indicating that our method is highly sensitive to boundaries between land features. Fig. 6 displays the confusion matrix derived from the classification results of various methods. Observations are revealed as follows.

1) By improving the construction of adjacency graphs and deeply exploring point-to-point structural relationships, misclassifications of geometrically distinct categories, such as road, building, and tree, are significantly reduced, as seen with GACNet and MKGSL.
2) Enhancing the use of spectral information markedly reduces misclassifications of less geometrically structured features such as water and powerline, as with DSGCN-ASR.

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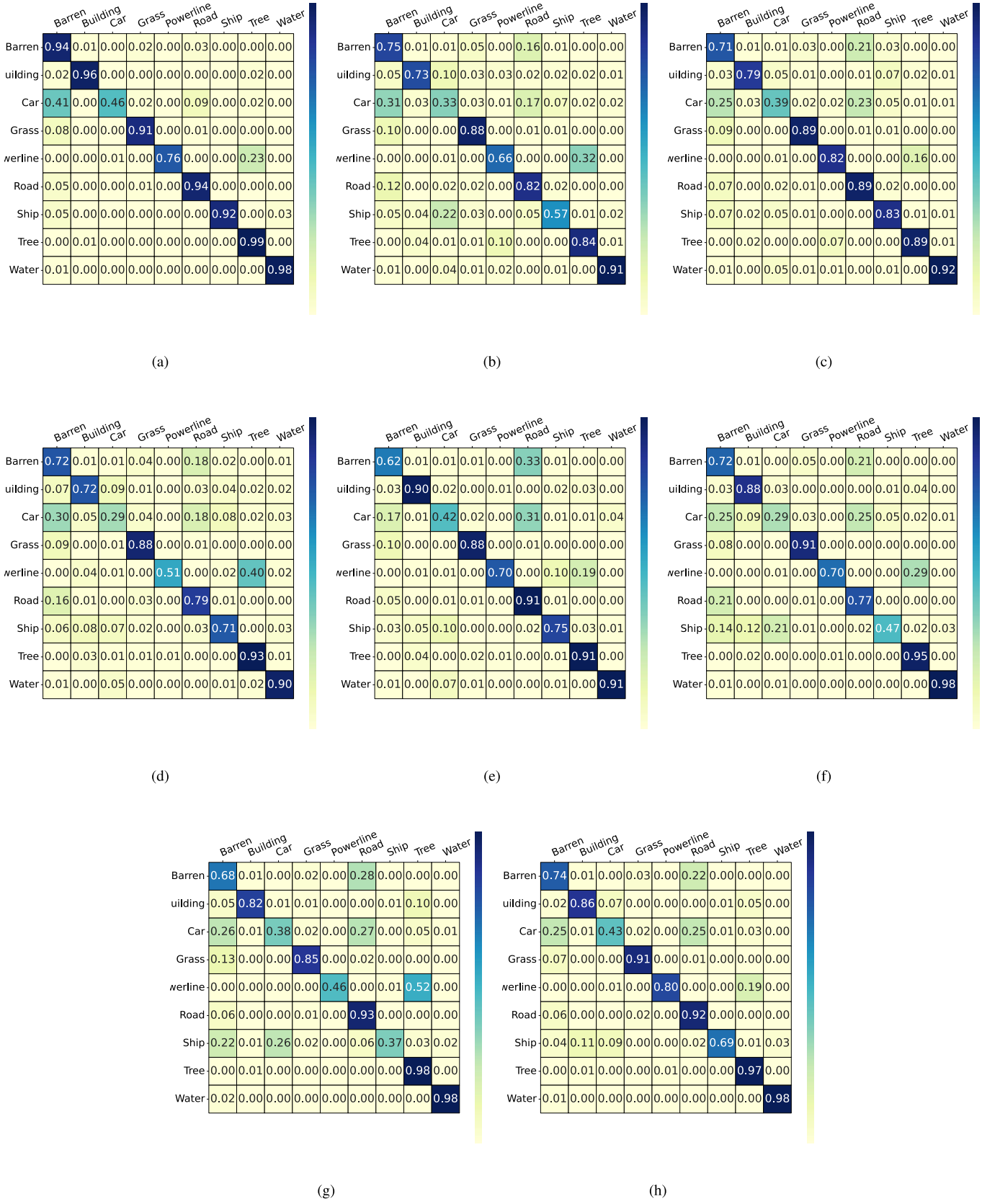


Fig. 6. Visualization of the confusion matrix for classification results on the HT dataset. (a) Ground truth. (b) GCN. (c) GCNII. (d) GACNet. (e) GCBNet. (f) DSGCN-ASR. (g) MKGSL. (h) MGC2N (Ours).

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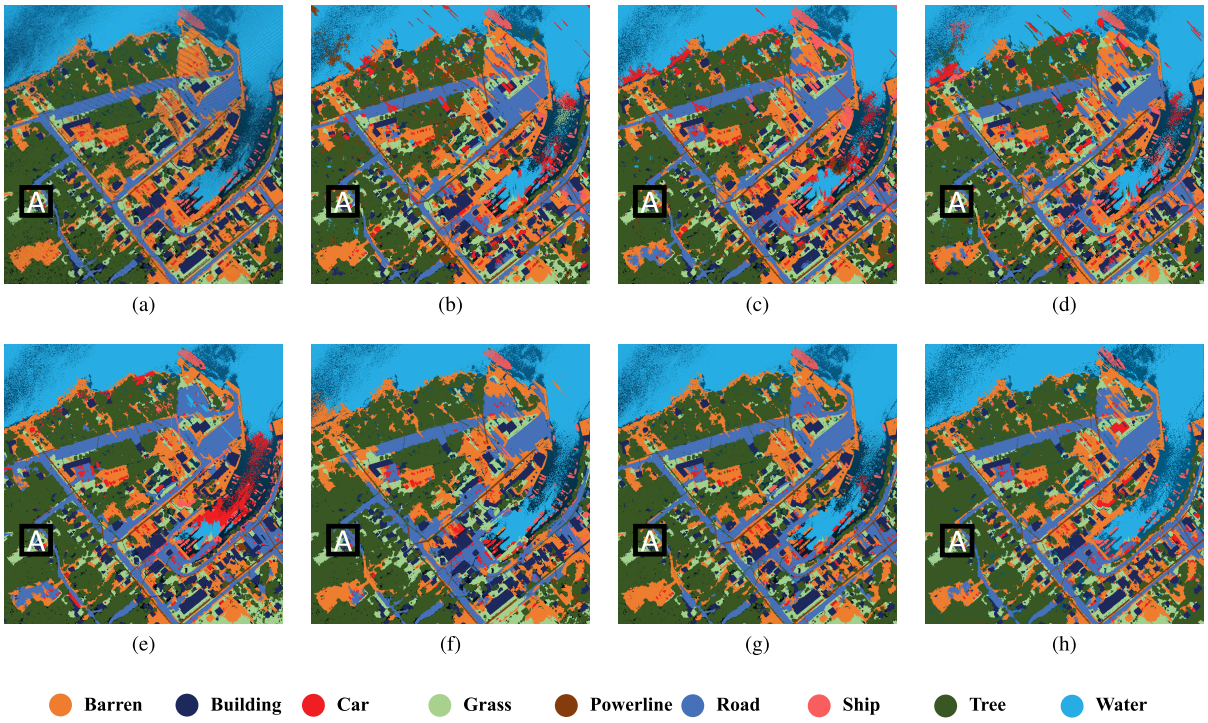


Fig. 7. Visualization of classification results on the HT dataset. (a) Ground truth. (b) GCN. (c) GCNII. (d) GACNet. (e) GCBNet. (f) DSGCN-ASR. (g) MKGSL. (h) MGC2N (Ours).

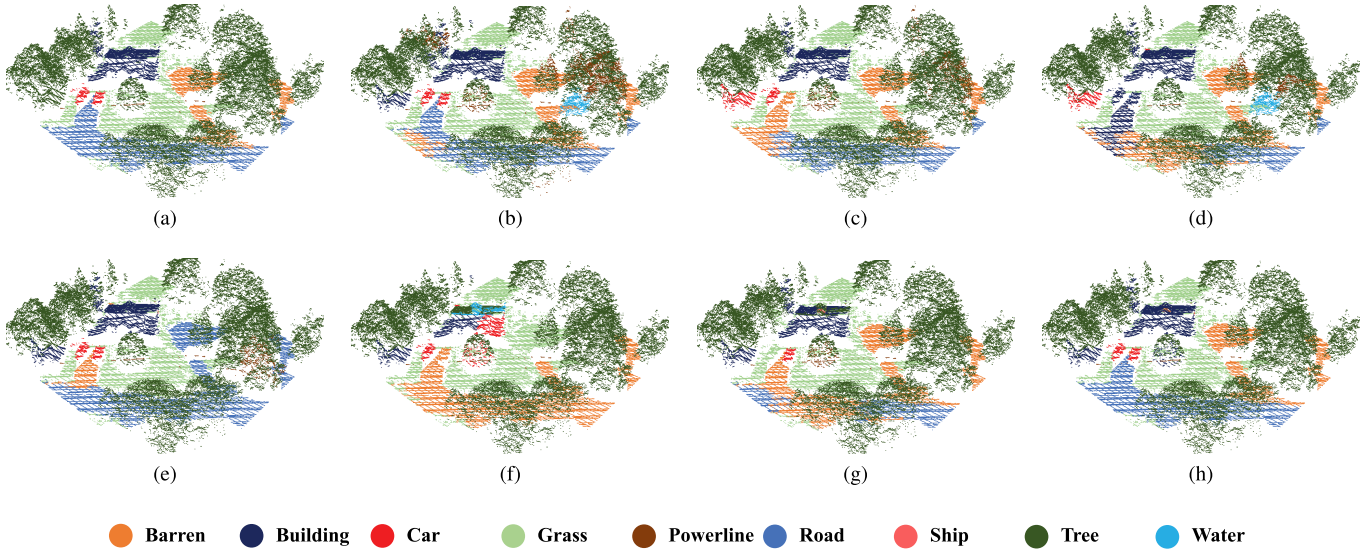


Fig. 8. Three-dimensional presentation of A on the HT dataset. (a) Ground truth. (b) GCN. (c) GCNII. (d) GACNet. (e) GCBNet. (f) DSGCN-ASR. (g) MKGSL. (h) MGC2N (Ours).

3) Methods that generally improve network inference capabilities without bias enhance overall network performance, such as GCNII and GCBNet. Our method adapts the construction of adjacency graphs to extract local geometric features from point clouds and employs cross-convolution to learn spectral differences between land features. In addition, we incorporate deep networks to strengthen model inference capabilities. The application of these integrated techniques likely contributes to the superior land feature recognition capabilities of our method.

The visualization results of different methods are shown in Fig. 7. Overall, our method is more consistent with the ground truth, and the geometric boundaries between land covers are more accurate and clearer. To illustrate the results of each method more intuitively, we selected area A for 3-D display, as depicted in Fig. 8. In contrast, our method excels in accurately recognizing various types of land covers. GCN, GCNII, GACNet, and GCBNet incorrectly classify a large number of points of tree as powerline, with GCN and GACNet even misclassifying them as water. In addition, GCN, GACNet, DSGCN-ASR, and MKGSL fail to identify cars in the area.

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TABLE II
COMPARISON OF EVALUATION METRICS (%) ON THE UH DATASET

Method	Metrics	Barren	Car	Commercial	Grass	Road	Powerline	Residential	Tree	Macro Average	OA
GCN [35]	Precision	54.13	36.22	70.84	81.62	76.47	70.78	74.60	77.39	67.76	67.39
	Recall	80.05	15.41	49.31	74.47	51.76	16.41	51.94	95.05	54.30	
	F-score	64.59	21.63	58.14	77.88	61.73	26.64	61.24	85.32	60.29	
	IoU	47.70	12.12	40.99	63.77	44.65	15.37	44.13	74.39	42.89	
GCNII [38]	Precision	44.95	45.80	81.14	84.53	85.18	**81.93**	77.36	81.10	72.75	66.80
	Recall	83.79	13.00	59.87	73.33	49.13	22.34	68.81	**96.07**	58.29	
	F-score	58.51	20.26	68.90	78.53	62.31	35.10	72.83	87.95	64.72	
	IoU	41.35	11.27	52.56	64.65	45.26	21.29	57.27	78.50	46.52	
GACNet [37]	Precision	38.88	55.28	59.79	80.38	77.53	62.68	79.12	76.85	66.31	61.31
	Recall	80.00	12.65	31.70	74.45	47.13	20.40	46.77	93.85	50.87	
	F-score	52.33	20.59	41.43	77.30	58.62	30.78	58.79	84.50	57.57	
	IoU	35.44	11.48	26.13	63.00	41.47	18.19	41.63	73.16	38.81	
GCBNet [39]	Precision	46.94	**60.35**	80.20	87.11	**85.25**	72.86	83.07	86.28	75.26	69.47
	Recall	**85.38**	20.16	71.41	78.00	46.53	30.04	71.46	95.97	62.37	
	F-score	60.58	30.22	75.55	82.30	60.20	42.54	76.83	90.87	68.21	
	IoU	43.45	17.80	60.71	69.93	43.06	27.02	62.37	83.26	50.95	
DSGCN-ASR [18]	Precision	**75.78**	33.89	67.28	83.02	72.70	79.49	83.82	88.28	73.03	78.20
	Recall	80.72	28.60	71.00	80.46	68.65	32.39	67.05	94.42	65.41	
	F-score	78.18	31.02	69.09	81.72	70.62	46.02	74.51	91.24	69.01	
	IoU	64.17	18.36	52.77	69.09	54.58	29.89	59.37	83.90	54.02	
MKGSL [19]	Precision	75.32	20.85	75.10	**88.16**	75.57	40.90	82.94	92.70	68.94	79.74
	Recall	83.17	**41.80**	**83.75**	74.78	67.55	29.04	**80.10**	92.84	69.13	
	F-score	**79.05**	27.82	79.19	80.92	71.33	33.96	**81.50**	92.77	69.04	
	IoU	**65.36**	16.16	65.55	67.96	55.44	20.46	**68.77**	86.52	55.78	
MGC2N(Ours)	Precision	73.60	42.92	**94.62**	84.95	74.99	79.33	**89.87**	**92.98**	**79.16**	**80.01**
	Recall	82.95	30.81	77.80	**80.62**	**70.51**	**61.58**	64.36	95.34	**70.50**	
	F-score	77.99	**35.87**	**85.39**	**82.73**	**72.68**	**69.34**	75.00	**94.15**	**74.58**	
	IoU	63.93	**21.85**	**74.50**	**70.55**	**57.09**	**53.06**	60.00	**88.94**	**61.24**	

DSGCN-ASR also misclassifies buildings into other classes. Furthermore, these methods are prone to confusing similar land covers, such as road and barren. In contrast, our method yields results that are more consistent with the ground truth. We are not only able to accurately distinguish between similar land covers but also effectively capture the geometries and boundaries of land covers.

2) Classification Results on UH: Table II shows the overall average metrics and the metrics for each class on the UH dataset. Due to the poor labeling quality of the UH dataset, all methods perform poorly on this dataset. In particular, the labeling of car classes is very confusing, which greatly affects the overall performance of the network. Our method achieved an OA of 80.01%, MP of 79.16%, MR of 70.50%, MF of 74.58%, and MIoU of 61.24%. Notably, MF achieved a significant improvement of 5.54% and MIoU saw a substantial increase of 5.46%. These enhancements indicate that our method recognizes the land covers more accurately. This is further evidenced by improvements across most classes, with particularly notable gains in recognizing Powerline, where we achieved an improvement of over ten percentage points. Fig. 9 shows the confusion matrix of the various methods on the UH dataset. Again, it can be seen that the advantages of each method are consistent with their performance on the HT dataset. However, it is possible that the utilization of the spectral information on the UH dataset is significantly weaker due to its lower quality. Meanwhile, due to the sample distribution and hyperpoint segmentation, a large number of cars are classified as barren, which affects the feature distribution between the two and affects the performance of the network. These may be the reasons why the improvement of our method on the UH dataset is not as obvious as on the HT dataset.

The visualization results of UH are depicted in Fig. 10. Overall, the consistency of our method with the ground truth is also relatively high. Similarly, in order to better present the visualization results, we chose a region for 3-D presentation, labeled B. As shown in Fig. 11, it can be seen that methods, such as GCN, GCNII, GACNet, and GCBNet, are heavily influenced by the quality of the dataset, misclassifying many of the other classes into car, resulting in a large number of confuse. Also, the DSGCN-ASR and MKGSL have poor discrimination between grass and barren, classifying a large number of barren as grass. In contrast, our method not only performs very well in both aspects but also improves the ability to discriminate between the two classes of land covers with mixed distributions of powerline and tree.

3) Discussion and Analysis: By combining the results from both datasets, our method demonstrates a significant overall performance enhancement. We achieve a 4% and 5% point improvement in both MF and MIoU, along with a 1% and 2% in OA. The modest improvement in OA can be attributed to class imbalance. While significant progress has been made in most classes, some with abundant instances, such as water and

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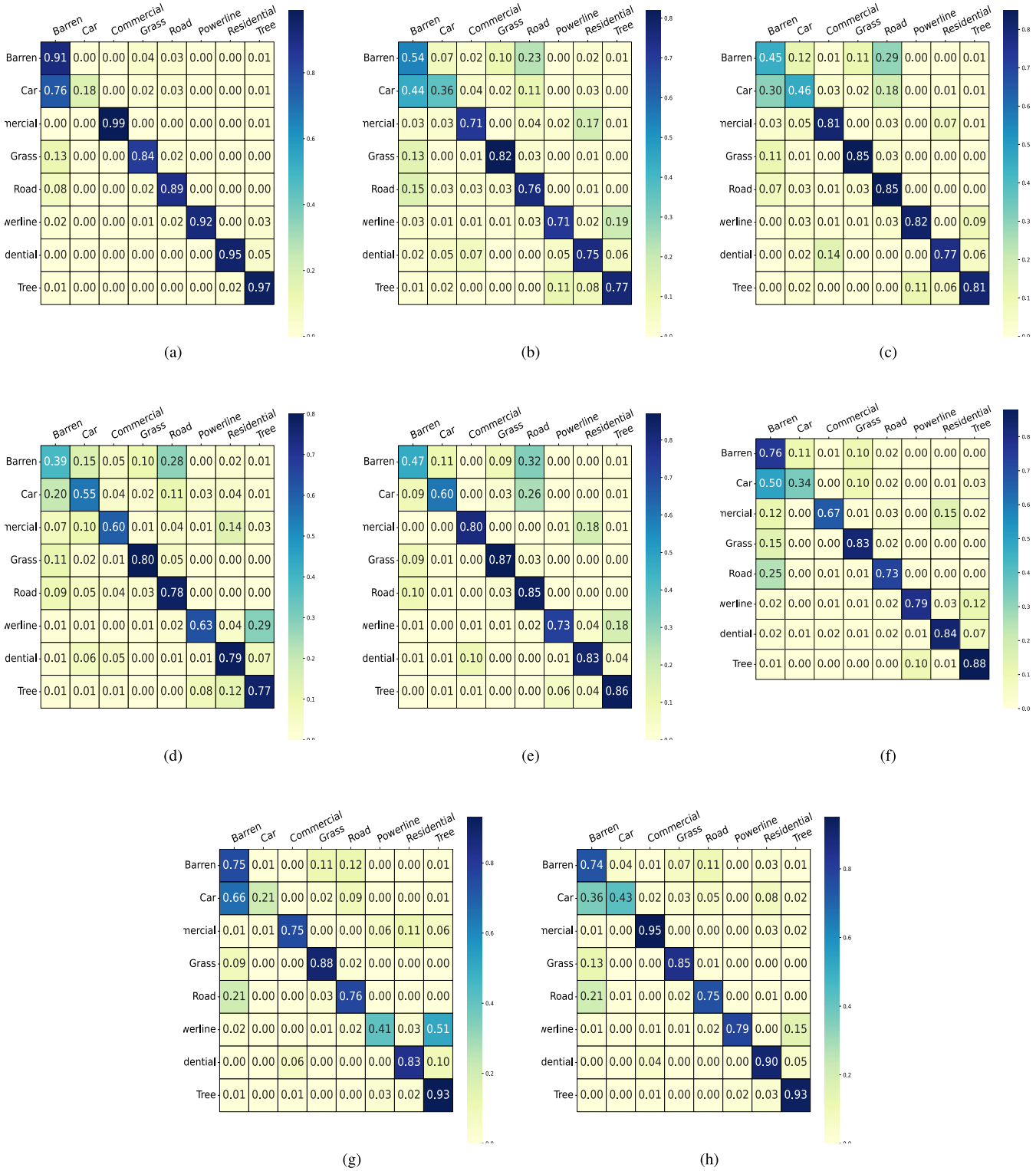


Fig. 9. Visualization of the confusion matrix for classification results on the UH dataset. (a) Ground truth. (b) GCN. (c) GCNII. (d) GACNet. (e) GCBNet. (f) DSGCN-ASR. (g) MKGSL. (h) MGC2N (Ours).

barren, have seen less evident improvements. Consequently, the enhancement in OA is not as pronounced as in other metrics. We analyzed the reasons for the poor performance of our method in these particular classes. Due to the limitations of data collection equipment and the inherent properties of water, some points exhibit distortion, leading to a bottleneck in the classification performance for the water class. In addition, the HT dataset has a limited number of car instances, and the impact of superpoint segmentation significantly affects our classification performance for this category. In the UH dataset,

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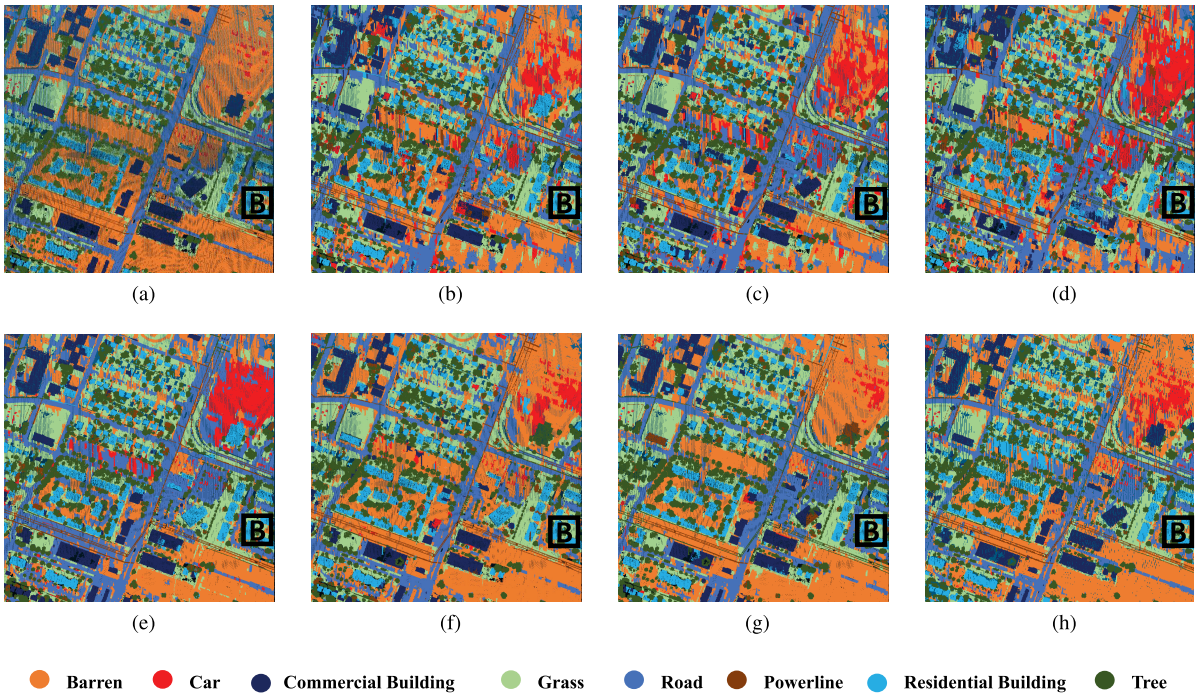


Fig. 10. Visualization of classification results on the UH dataset. (a) Ground truth for superpoint mapping. (b) GCN. (c) GCNII. (d) GACNet. (e) GCBNet. (f) DSGCN-ASR. (g) MKGSL. (h) MGC2N (Ours).

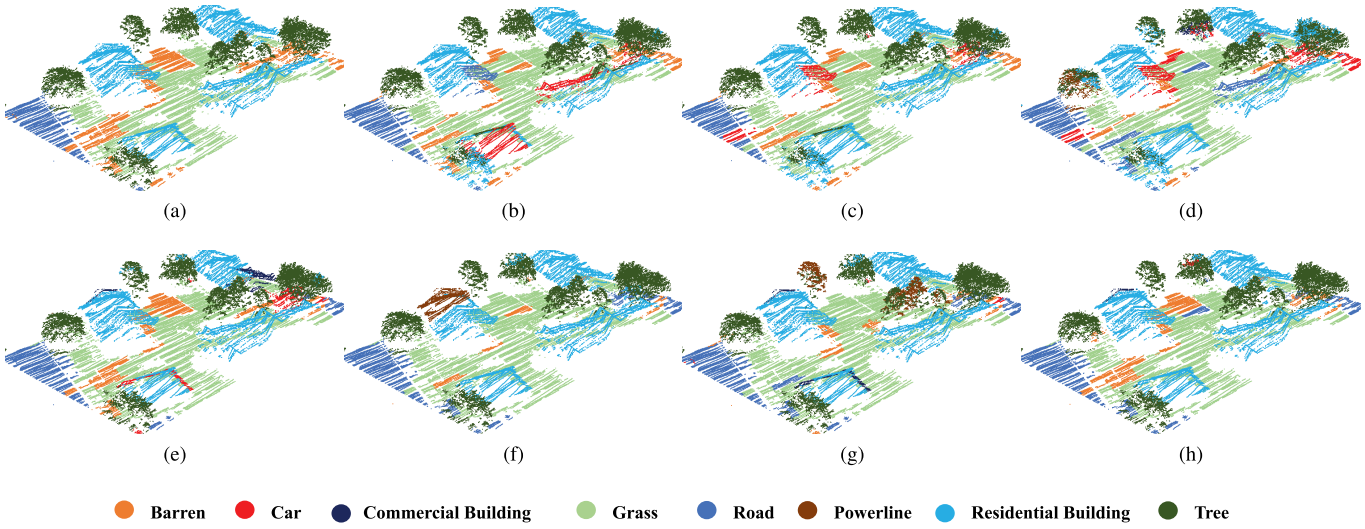


Fig. 11. Three-dimensional presentation of B on the UH dataset. (a) Ground truth. (b) GCN. (c) GCNII. (d) GACNet. (e) GCBNet. (f) DSGCN-ASR. (g) MKGSL. (h) MGC2N.

the extensive distribution and irregular geometric shapes, as well as the significant confusion between the barren and road annotations, pose challenges for accurately recognizing the barren class using our method.

Overall, our method significantly enhances the performance of multispectral point cloud classification. The improved visualization results clearly indicate that our approach is more accurate in recognizing geometric structures and delineating land cover boundaries. These comprehensive experimental results affirm the effectiveness and robustness of our method in capturing detailed spatial information for precise classification.

TABLE III
ABLATION EXPERIMENT SETTING

Group	SAM	S2C2
I	✗	✗
II	✓	✗
III	✗	✓
IV	✓	✓

C. Ablation Experiments

To systematically evaluate the contributions of the two modules within our network, we designed four sets of

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TABLE IV
EVALUATION METRICS (%) FOR ABLATION EXPERIMENTS ON THE HT DATASET

Group	Metrics	Barren	Building	Car	Grass	Powerline	Road	Ship	Tree	Water	Macro Average	OA
I	Precision	66.02	66.98	35.73	89.33	75.12	52.03	49.24	90.02	87.13	67.95	79.34
	Recall	69.81	81.14	19.17	62.83	7.66	**71.37**	46.55	97.14	83.51	59.91	
	F1-score	67.86	73.38	24.96	73.77	13.90	60.18	47.86	93.44	85.28	63.68	
	IoU	51.36	57.96	14.26	58.44	7.47	43.04	31.46	87.69	74.33	47.33	
II	Precision	**75.25**	**92.66**	13.53	87.65	75.46	78.32	68.51	95.92	60.57	71.99	86.35
	Recall	82.15	66.05	25.33	**95.44**	19.57	68.74	39.68	97.64	95.57	65.58	
	F1-score	78.55	77.13	17.64	**91.38**	31.08	73.22	50.26	96.78	74.15	68.63	
	IoU	64.68	62.77	9.67	**84.13**	18.40	57.75	33.56	93.75	58.92	53.74	
III	Precision	65.56	88.77	**44.03**	91.19	45.50	80.51	**73.84**	94.97	98.09	75.83	86.87
	Recall	82.64	78.46	**58.28**	73.92	16.68	64.96	60.27	98.84	**99.14**	70.35	
	F1-score	73.12	83.30	**50.16**	81.65	24.41	71.90	66.37	96.87	**98.61**	72.99	
	IoU	57.62	71.37	**33.48**	68.99	13.90	56.13	49.67	93.92	**97.26**	60.26	
IV	Precision	73.53	86.05	43.09	**91.25**	**79.69**	**92.28**	68.93	**96.86**	**98.38**	**81.12**	**90.30**
	Recall	**88.00**	**84.09**	43.32	91.39	**39.99**	68.06	**70.97**	**99.04**	98.61	**75.94**	
	F1-score	**80.12**	**85.06**	43.20	91.32	**53.25**	**78.34**	**69.93**	**97.94**	98.49	**78.44**	
	IoU	**66.83**	**74.00**	27.55	84.03	**36.29**	**64.39**	**53.77**	**95.96**	97.03	**66.65**	

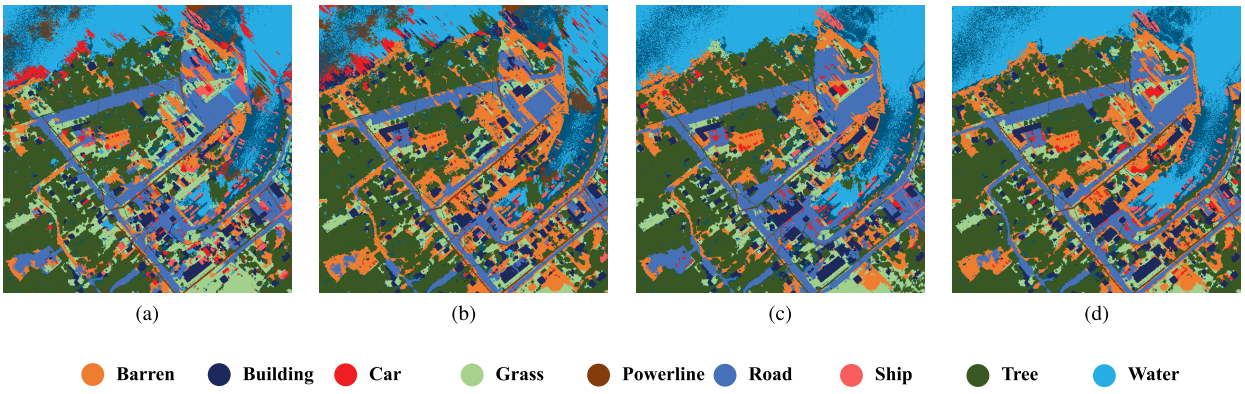


Fig. 12. Visualization results for ablation experiments on the HT dataset. (a) I. (b) II. (c) II. (d) IV.

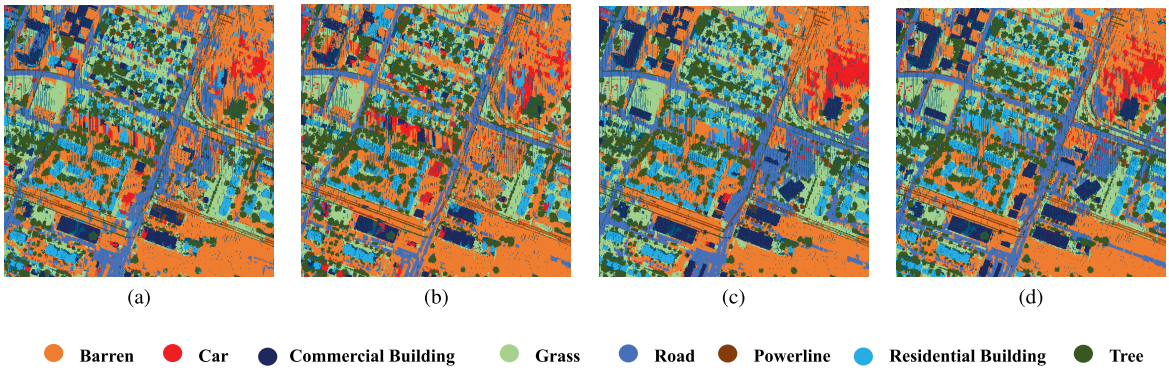


Fig. 13. Visualization results for ablation experiments on the UH dataset. (a) I. (b) II. (c) II. (d) IV.

experiments: I. Neither module is used—II: using only the SAM module; III: using only the S2C2 module; and IV: using both modules. Table III shows the relationship between the different experimental groups and the two modules. Tables IV and V quantitatively present the ablation experiment results for the HT and UH datasets, respectively, while Figs. 12 and 13 qualitatively show the results for HT and UH, respectively.

1) Effectiveness and Performance of the SAM Module: To dynamically acquire effective contextual information for each point, we designed the SAM module, which constructs an independent neighborhood for each point through training. As can be seen from the results of these experiments, the addition of the SAM module significantly enhances our feature extraction and perception capabilities. On the HT dataset, it achieved an improvement of 6.41% in MIoU and

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TABLE V
EVALUATION METRICS (%) FOR ABLATION EXPERIMENTS ON THE UH DATASET

Method	Metrics	Barren	Car	Commercial	Grass	Road	Powerline	Residential	Tree	Macro Average	OA
I	Precision	52.18	**54.90**	86.87	84.71	73.68	**86.76**	71.64	62.48	71.65	64.66
	Recall	78.94	22.50	51.96	74.50	49.70	8.81	**76.37**	94.34	57.14	
	F1-score	62.83	31.92	65.03	79.28	59.36	15.99	73.93	75.17	63.58	
	IoU	45.80	18.99	48.18	65.67	42.21	8.69	58.64	60.22	43.55	
II	Precision	63.44	48.70	75.31	83.41	69.96	74.13	81.69	90.40	73.38	73.67
	Recall	80.02	18.50	43.96	**81.39**	63.43	38.77	58.38	**95.56**	60.00	
	F1-score	70.77	26.82	55.52	82.39	66.54	50.92	68.10	92.91	66.02	
	IoU	54.77	15.49	38.42	70.05	49.86	34.15	51.63	86.76	50.14	
III	Precision	61.79	39.12	87.60	80.74	**83.20**	35.00	79.59	91.46	69.81	74.44
	Recall	82.01	**33.57**	**78.12**	77.45	52.24	42.12	71.70	93.15	66.29	
	F1-score	70.48	**36.13**	82.59	79.06	64.19	38.23	**75.44**	92.29	68.01	
	IoU	54.41	**22.05**	70.35	65.37	47.26	23.63	**60.57**	85.69	53.67	
IV	Precision	**73.60**	42.92	**94.62**	**84.95**	74.99	79.33	**89.87**	**92.98**	**79.16**	**80.01**
	Recall	**82.95**	30.81	77.80	80.62	**70.51**	**61.58**	64.36	95.34	**70.50**	
	F1-score	**77.99**	35.87	**85.39**	**82.73**	**72.68**	**69.34**	75.00	**94.15**	**74.58**	
	IoU	**63.93**	21.85	**74.50**	**70.55**	**57.09**	**53.06**	60.00	**88.94**	**61.24**	

7.01% in OA, as shown in Table IV. On the UH dataset, the improvements were 6.59% in MIoU and 9.01% in OA, as shown in Table V. The visualized results also demonstrate that the inclusion of the SAM module leads to more precise recognition and segmentation of land covers, as shown in Figs. 12 and 13.

2) Effectiveness and Performance of the S2C2 Module: To improve the network's ability to capture land cover geometry and boundaries, we designed the S2C2 module to enhance the diversity and effectiveness of global and local features. Similarly, the inclusion of the S2C2 module significantly improves the network's ability to classify multispectral point clouds. On the HT dataset, it achieved an improvement of 12.93% in MIoU and 7.53% in OA. On the UH dataset, the improvements were 10.12% in MIoU and 9.78% in OA. Notably, it is able to achieve a double-digit increase in MIoU, which demonstrates the ability of the S2C2 module to capture the spatial geometry of the land covers. Figs. 12 and 13 clearly demonstrate that incorporating the S2C2 module results in more accurate land cover segmentation.

By integrating the SAM and S2C2 modules, we have achieved a significant improvement in our network's performance, yielding the best classification results. This fully demonstrates the effectiveness and complementarity of the two modules, validating the advancement of our methodology.

V. CONCLUSION

In this article, we introduce MGC2N, a novel network that thoroughly explores the synergistic relationship between spectral and spatial features in multispectral point clouds. MGC2N dynamically learns representative contextual information for each point using a masking strategy, thereby achieving efficient processing of multispectral point clouds. We employ a cross-convolution approach to extract pairs of more efficient global and local features, enhancing the ability to capture spatial geometric features and land cover boundaries. In addition, cross-convolution reduces the similarity of higher order semantic features. We have conducted extensive experiments with SOTA methods and have demonstrated the superiority of our method. MGC2N achieves SOTA classification accuracies on two real multispectral point cloud datasets, with 90.38% OA and 66.65% MIoU on the HT dataset, and 80.01% OA and 61.24% MIoU on the UH dataset. The visualization results also demonstrate that our method can extract more detailed spatial features and achieve more accurate boundary segmentation.

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